Francesco Cavina

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Data & AI Platform Engineer with a BSc in Econometrics and Data Science from VU Amsterdam and incoming MSc studies in Econometrics, Operations Research and Business Analytics. Experience building privacy-aware AI workflows, data pipelines, local LLM infrastructure, RAG prototypes and full-stack tools with Python, FastAPI, TypeScript and cloud infrastructure. Interested in reliable LLM systems for adaptive learning, retrieval pipelines, content models and user-facing AI products.

TECHNICAL SKILLS

- **AI / LLM Engineering:** LLM workflows, RAG, vector search, ChromaDB, OpenAI API, Hugging Face, prompt design, structured outputs, citation-based generation, AI-assisted data discovery, hallucination-aware evaluation
- **Backend & APIs:** Python, FastAPI, SQL, REST APIs, SQLAlchemy, data pipelines, metadata-driven ingestion, schema inference, data quality workflows
- **Frontend & Product:** TypeScript, JavaScript, Next.js, React, Tailwind CSS, UI components, self-service tools, user-facing prototypes, technical demos
- **Data & Infrastructure:** PySpark, Databricks, Airflow, Docker, Kubernetes, Terraform, Azure, Git, CI/CD concepts
- **Relevant Strengths:** LLM-safe data access, modular platform thinking, documentation, technical communication, educational explanation, translating unclear workflows into working prototypes

RELEVANT EXPERIENCE

Data & AI Platform Engineer

Robodata — **September 2025 - June 2026, Amsterdam, Netherlands**

- Built privacy-aware AI and data platform components for enterprise clients, combining metadata-driven ingestion, schema inference, PII classification and governed access to internal data.
- Developed automated Python, PySpark and SQL pipelines using Bronze/Silver/Gold architecture, with attention to maintainability, data quality and downstream AI readiness.
- Built local LLM workflows for data discovery, field classification and metadata generation, helping transform raw enterprise datasets into structured, AI-usable knowledge assets.
- Created AI-ready data interfaces that allowed governed datasets to be queried safely and consistently by LLM-based tools and agents.
- Worked on self-service data tooling so non-specialist users could understand, operate and reuse technical data infrastructure without depending on engineers for every workflow.
- Documented platform components, workflow logic and design choices to support handover, adoption and future scaling.
- Used Docker, Kubernetes, Terraform, Azure, Databricks, Airflow and SQLAlchemy across production platform work.

Operational Trader

QuantFi — **October 2022 - May 2023, Amsterdam, Netherlands**

- Developed and tested Python based trading strategies using Random Forest, XGBoost, neural networks and walk forward validation.
- Built data ingestion, retraining and daily forecast workflows in Python and SQL to support recurring trading decisions.
- Created dashboards and visualisations to compare model signals, realised performance and forecast quality.
- Evaluated model output with attention to overfitting, changing market conditions and practical usability.
- Worked in an environment where speed, clarity and disciplined decision making mattered daily.

Teaching Assistant

Vrije Universiteit Amsterdam — **June 2022 - November 2024, Amsterdam, Netherlands**

- Taught weekly Probability Theory tutorials to first-year Econometrics students, translating abstract mathematical concepts into clear, structured explanations.
- Helped students move from memorising procedures to understanding assumptions, reasoning steps and problem-solving strategies.
- Built a student portfolio system for first-year students, combining educational support with practical tooling.
- Represented the programme during Open Days, explaining the curriculum and student experience to prospective students.

EDUCATION

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 • Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds • Leeds, England • 2024 • 8.5/10

PROJECTS

1) AI Powered Personal Website • https://github.com/FrancescoCavina02/My-Website • November 2025 - present
   Next.js, TypeScript, FastAPI, Python, Tailwind CSS
   Built a full-stack portfolio with a FastAPI backend that parses an Obsidian vault into structured API endpoints for notes, quotes and chat functionality. Designed it as a self-service knowledge interface, combining content modelling, backend automation, frontend usability and LLM-based interaction.

2) RAG Knowledge Chatbot • https://github.com/FrancescoCavina02/Spiritual-chatbot • November 2025 - present
   Python, FastAPI, ChromaDB, sentence transformers, LLM APIs
   Built a retrieval-augmented chatbot using semantic chunking, vector search, structured prompts and citation-based responses. Focused on reliable retrieval, traceable answers, hallucination reduction and reusable workflow design. Designed the system around the core problem of making a knowledge base queryable by an LLM while preserving source grounding.

3) Dodgeball Club Amsterdam • https://github.com/FrancescoCavina02/DCA-website • June 2025 - present
   HTML, CSS, JavaScript
   Built and maintain a public website focused on clear navigation, practical usability and simple content management for club members.

Podcasts

Podcast Link - Created two independent interview podcasts with researchers in physics, cosmology, mathematics and philosophy.

RELEVANT FIT FOR TECH

- Practical experience building LLM/data platform components, not only experimenting with AI tools.
- Strong overlap with RAG, AI-ready data interfaces, backend APIs, full-stack prototypes and cloud-native deployment.
- Educational experience as a Probability Theory teaching assistant, with a strong interest in how complex knowledge can be structured, explained and personalised.
- Comfortable working across AI engineering, software development, documentation and user-facing tooling.